

## AHWAZ, Iran, Islamic Republic of

WMO: 408110

Lat: 31.3443N

Lon: 48.7440E

Elev: 72

StdP: 14.66

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 40.8       | 43.1 | 21.1                          | 15.8 | 70.3 | 24.8 | 19.0 | 68.7 | 21.2                  | 63.5 | 18.2 | 63.6 | 3.0                   | 280  | 0.443 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |       |      | Evaporation WB/MCDB |      |      |      |      |       | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|-------|------|---------------------|------|------|------|------|-------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%    |      | 0.4%                |      | 1%   |      | 2%   |       |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB    | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB  | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)   | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)   | (o)                  | (p)  |
| 7             | 28.0                   | 118.5           | 74.3 | 116.7 | 73.9 | 114.9 | 73.2 | 83.8                | 98.1 | 81.3 | 99.8 | 79.0 | 102.7 | 7.5                  | 290  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |       | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|-------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |       |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB  |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)   |                   |
| 80.6                            | 159.3 | 92.5 | 76.8 | 140.0 | 92.7 | 71.9 | 118.1 | 94.7 | 48.2          | 96.0 | 45.3 | 98.6 | 42.7 | 101.9 | 92.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 18.7              | 15.8 | 13.6 | 36.0                       | 121.9 | 3.1                | 2.1 | 33.8                                               | 123.4 | 32.0       | 124.6 | 30.3       | 125.8 | 28.0       | 127.3 |
|                   |      |      | 33.1                       | 86.4  | 3.1                | 2.9 | 30.9                                               | 88.5  | 29.0       | 90.1  | 27.3       | 91.8  | 25.0       | 93.8  |
|                   |      |      | DB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |      | WB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 79.7   | 55.7  | 60.1  | 68.3  | 78.4  | 90.6  | 98.0  | 101.3 | 100.5 | 93.1  | 83.1  | 67.9  | 58.1 |
|                                                                                  | DBStd       | 17.14  | 4.94  | 5.36  | 5.46  | 5.85  | 5.01  | 3.37  | 2.70  | 2.79  | 4.14  | 5.33  | 6.25  | 5.18 |
|                                                                                  | HDD50       | 19     | 11    | 3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 5    |
|                                                                                  | HDD65       | 721    | 289   | 149   | 26    | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 39    | 217  |
|                                                                                  | CDD50       | 10867  | 189   | 286   | 569   | 853   | 1260  | 1440  | 1591  | 1566  | 1294  | 1026  | 536   | 257  |
|                                                                                  | CDD65       | 6096   | 1     | 12    | 130   | 405   | 795   | 990   | 1126  | 1101  | 844   | 561   | 126   | 5    |
|                                                                                  | CDH74       | 99835  | 5     | 88    | 1245  | 4888  | 13075 | 17896 | 20765 | 19835 | 13784 | 7143  | 1070  | 41   |
|                                                                                  | CDH80       | 71503  | 0     | 5     | 407   | 2531  | 8982  | 13598 | 16301 | 15373 | 9758  | 4209  | 338   | 1    |
| Wind                                                                             | WSAvg       | 5.3    | 4.3   | 4.9   | 5.3   | 5.6   | 6.0   | 7.5   | 7.3   | 6.1   | 5.0   | 4.0   | 3.8   | 3.9  |
| Precipitation                                                                    | PrecAvg     | 8.1    | 1.7   | 0.9   | 1.1   | 0.8   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.3   | 1.3   | 1.8  |
|                                                                                  | PrecMax     | 16.3   | 4.6   | 2.9   | 4.9   | 3.7   | 1.7   | 0.0   | 0.0   | 0.0   | 0.2   | 2.4   | 6.8   | 5.2  |
|                                                                                  | PrecMin     | 3.1    | 0.0   | 0.0   | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0  |
|                                                                                  | PrecStd     | 3.1    | 1.2   | 0.8   | 1.1   | 0.7   | 0.3   | 0.0   | 0.0   | 0.0   | 0.0   | 0.6   | 1.5   | 1.6  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 73.7  | 80.4  | 94.7  | 103.9 | 114.4 | 119.8 | 121.8 | 120.1 | 115.2 | 106.2 | 91.5  | 78.5 |
|                                                                                  |             | MCWB   | 57.5  | 59.6  | 61.5  | 66.4  | 71.4  | 73.4  | 75.3  | 74.2  | 73.0  | 69.0  | 67.1  | 60.1 |
|                                                                                  | 2%          | DB     | 70.2  | 76.8  | 88.1  | 99.0  | 110.9 | 116.8 | 118.7 | 118.3 | 113.0 | 103.7 | 87.8  | 74.1 |
|                                                                                  |             | MCWB   | 57.2  | 58.4  | 61.5  | 65.8  | 70.2  | 72.4  | 74.5  | 74.6  | 72.7  | 69.8  | 64.9  | 59.6 |
|                                                                                  | 5%          | DB     | 67.9  | 73.6  | 84.2  | 95.9  | 107.8 | 114.8 | 116.8 | 116.5 | 110.3 | 100.6 | 84.1  | 71.2 |
|                                                                                  |             | MCWB   | 56.4  | 57.5  | 60.5  | 64.6  | 68.9  | 71.5  | 74.0  | 74.5  | 71.7  | 68.5  | 63.9  | 59.5 |
|                                                                                  | 10%         | DB     | 64.8  | 71.3  | 80.6  | 92.8  | 105.5 | 112.8 | 114.9 | 114.6 | 107.8 | 97.2  | 80.4  | 68.0 |
|                                                                                  |             | MCWB   | 55.5  | 56.4  | 59.3  | 64.1  | 68.5  | 70.6  | 73.2  | 74.2  | 70.6  | 67.6  | 62.2  | 58.2 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 63.3  | 64.6  | 68.1  | 71.6  | 75.9  | 80.3  | 85.0  | 87.5  | 84.4  | 79.6  | 72.6  | 66.0 |
|                                                                                  |             | MCDB   | 68.0  | 72.5  | 77.8  | 88.6  | 104.4 | 104.3 | 100.9 | 97.6  | 93.0  | 89.6  | 83.1  | 70.9 |
|                                                                                  | 2%          | WB     | 60.9  | 62.3  | 65.6  | 69.3  | 73.6  | 77.3  | 81.9  | 84.8  | 81.3  | 76.1  | 69.1  | 63.6 |
|                                                                                  |             | MCDB   | 66.2  | 70.0  | 77.5  | 87.7  | 102.0 | 104.5 | 102.2 | 96.8  | 94.8  | 89.8  | 80.2  | 69.2 |
|                                                                                  | 5%          | WB     | 59.2  | 60.2  | 63.7  | 67.9  | 71.9  | 74.7  | 79.4  | 82.5  | 78.3  | 73.5  | 66.9  | 61.4 |
|                                                                                  |             | MCDB   | 64.5  | 68.6  | 76.5  | 86.5  | 99.3  | 106.8 | 105.4 | 99.8  | 98.3  | 89.0  | 78.0  | 67.3 |
|                                                                                  | 10%         | WB     | 57.1  | 58.5  | 61.9  | 66.6  | 70.5  | 72.8  | 76.9  | 79.9  | 75.2  | 71.1  | 64.8  | 59.5 |
|                                                                                  |             | MCDB   | 63.3  | 67.3  | 75.7  | 85.3  | 98.0  | 107.0 | 106.9 | 101.8 | 99.3  | 89.8  | 75.2  | 65.6 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 17.0  | 19.4  | 22.3  | 23.8  | 26.3  | 29.1  | 28.0  | 29.2  | 30.2  | 27.3  | 21.6  | 17.8 |
|                                                                                  |             | MCDBR  | 20.5  | 22.8  | 26.9  | 27.7  | 30.0  | 31.6  | 30.2  | 30.6  | 31.4  | 30.5  | 25.9  | 22.4 |
|                                                                                  |             | MCWBR  | 10.9  | 10.2  | 9.6   | 8.3   | 9.5   | 10.3  | 10.2  | 11.4  | 11.6  | 11.1  | 11.1  | 11.4 |
|                                                                                  | 5% WB       | MCDBR  | 15.2  | 17.8  | 21.1  | 23.0  | 26.6  | 30.6  | 29.0  | 28.3  | 28.4  | 24.9  | 19.4  | 16.0 |
| MCWBR                                                                            |             | 10.2   | 10.0  | 9.8   | 8.8   | 9.9   | 11.6  | 12.5  | 13.9  | 14.4  | 13.5  | 10.8  | 10.1  |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.468  | 0.504 | 0.563 | 0.660 | 0.639 | 0.590 | 0.660 | 0.606 | 0.532 | 0.552 | 0.499 | 0.466 |      |
|                                                                                  | taud        | 1.917  | 1.835 | 1.697 | 1.560 | 1.596 | 1.660 | 1.561 | 1.655 | 1.785 | 1.762 | 1.872 | 1.938 |      |
|                                                                                  | Ebn at Noon | 228    | 234   | 231   | 215   | 221   | 232   | 215   | 225   | 236   | 217   | 216   | 220   |      |
|                                                                                  | Edh at Noon | 50     | 60    | 74    | 88    | 85    | 79    | 87    | 79    | 66    | 63    | 52    | 47    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 996    | 1305  | 1684  | 1895  | 2242  | 2568  | 2415  | 2271  | 1997  | 1533  | 1121  | 929   |      |
|                                                                                  | RadStd      | 74     | 94    | 87    | 119   | 110   | 72    | 59    | 63    | 62    | 69    | 86    | 88    |      |

## Historical Trends

|                        |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only           |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (7 neighbors) |  | +0.54 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page